TITANIC SURVIVOR ANALYSIS

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DATASET & OBJECTIVE

- What passenger characteristics were associated with survival on the Titanic?

- Google Colab (Python)
- df.info() = 891 passengers & 12 original variables

index	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.925	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05	NaN	S

DATA CLEANING AND FINDINGS

```
print(df.isnull().sum())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

Median age = 28.0
Why?

```
df[df['Embarked'].isnull()]
```

ıss	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	Icard, Miss. Amelie	female	38.0	0	0	113572	80.0	B28	NaN
1	Stone, Mrs. George Nelson (Martha Evelyn)	female	62.0	0	0	113572	80.0	B28	NaN

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

Mode = S
Why?

DATA CLEANING AND FINDINGS

Useful?

```
df['Cabin']
```

	Cabin
0	NaN
1	C85
2	NaN
3	C123
4	NaN
...	...
886	NaN
887	B42
888	NaN
889	C148
890	NaN

891 rows × 1 columns

```
df['Ticket']
```

	Ticket
0	A/5 21171
1	PC 17599
2	STON/O2. 3101282
3	113803
4	373450
...	...
886	211536
887	112053
888	W./C. 6607
889	111369
890	370376

891 rows × 1 columns

Consistency

```
df['Fare'] = df['Fare'].round(2)
```

```
df['Fare']
```

	Fare
0	7.25
1	71.28
2	7.92
3	53.10
4	8.05
...	...
886	13.00
887	30.00
888	23.45
889	30.00
890	7.75

891 rows × 1 columns

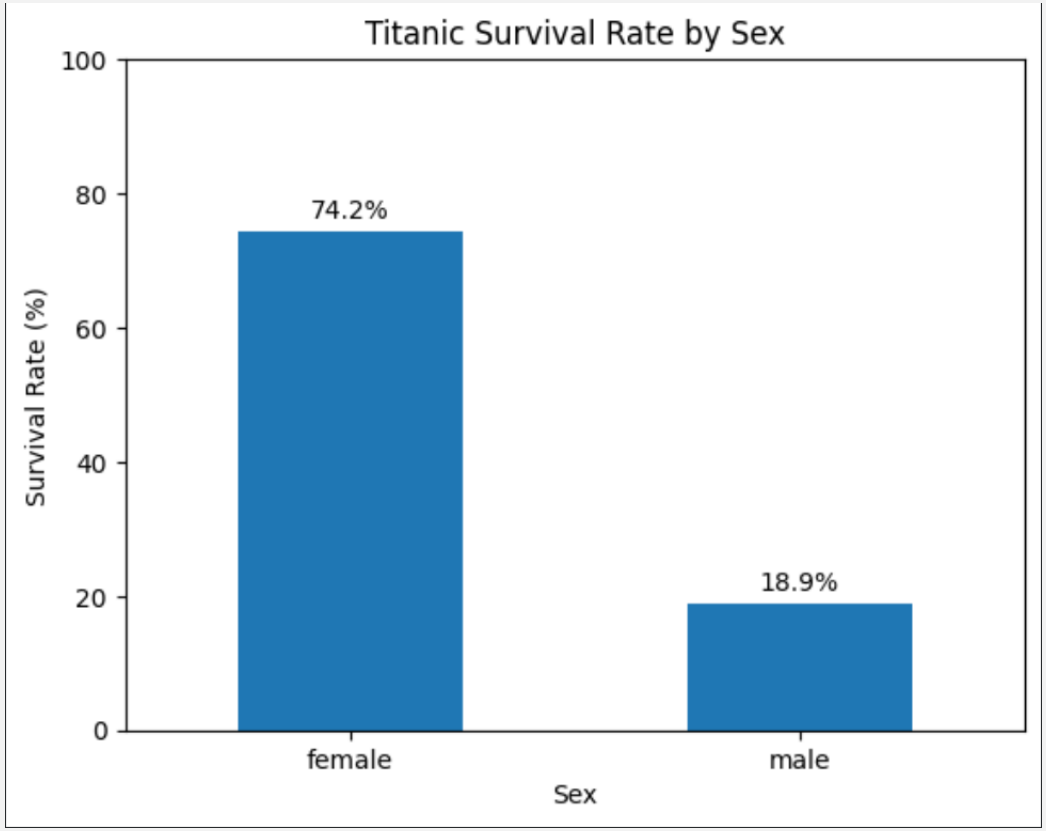
Outliers Findings

```
fare_outliers[['PassengerId', 'Survived', 'Fare']].sort_values(
    by='Fare',
    ascending=False
)
```

	PassengerId	Survived	Fare
258	259	1	512.33
737	738	1	512.33
679	680	1	512.33
27	28	0	263.00
341	342	1	263.00
...	...	...	...
324	325	0	69.55
369	370	1	69.30
641	642	1	69.30
336	337	0	66.60
151	152	1	66.60

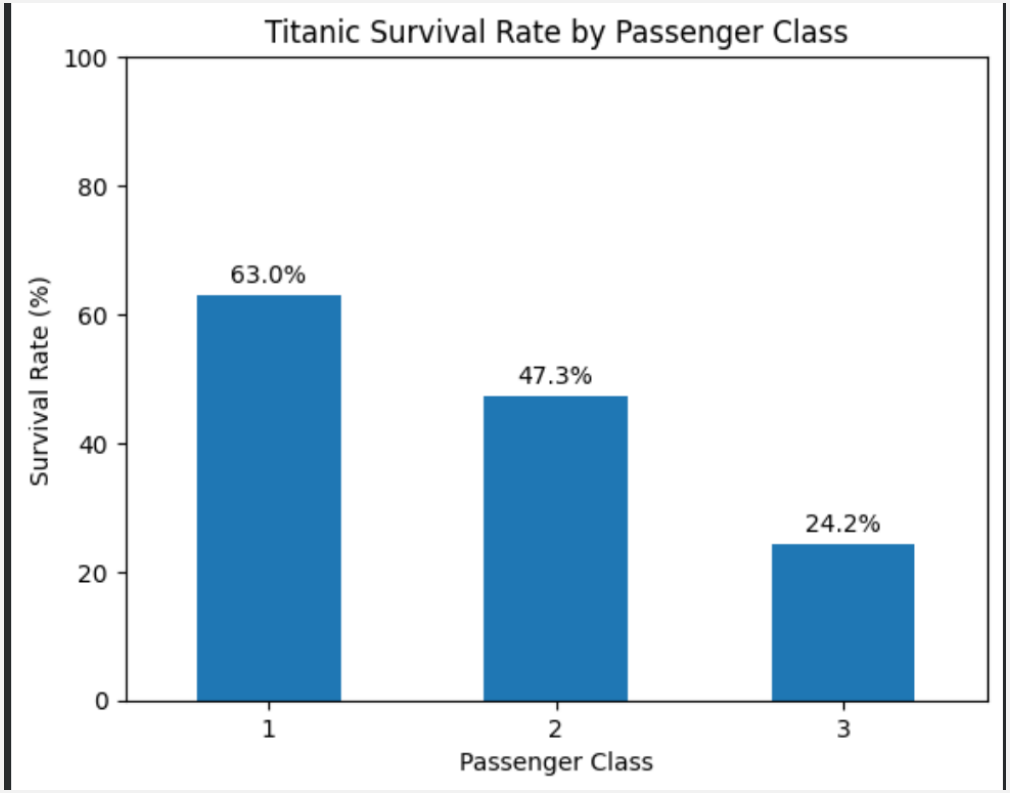
116 rows × 3 columns

SURVIVAL RATES BASED ON GENDER



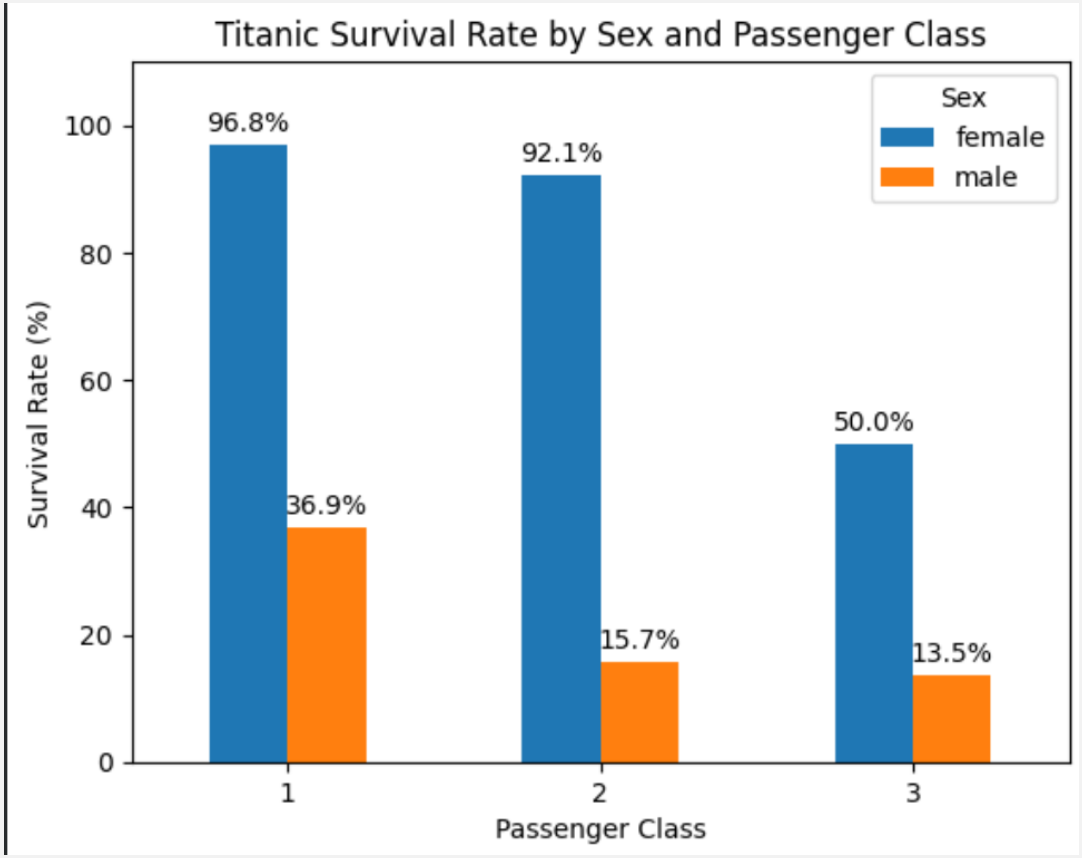
- Did survival differ from sex?
- Female vs Male survival rates
- Total Survivals: 342 out of 891
- About 74% of females survived vs 19% of males survived
- Females were more likely to live than males

SURVIVAL RATES BASED ON FARES



- Did survival differ by passenger class?
- Out of 891 total passengers:
  - Fare 1: 216
  - Fare 2: 184
  - Fare 3: 491
- The most expensive tickets had a higher rate of survival based on the accessibilities that they have.

SURVIVAL RATES BASED ON SEX AND PASSENGER CLASS



- Women had a dominant group in each field of fares
- Women were almost guaranteed in fare 1 and 2
- Men had a difficult time in each fare and it based on survival skills and possible luck

KEY FINDINGS

- Gender showed the largest difference in survival
  - 74.2% female vs 18.9% male
- Passenger class was also associated with survival:
  - Major decrease from 2nd to 3rd class
- Gender and class together showed a clear pattern
  - Women 1st class survival 96.8% vs 3rd class men 13.5%
  - Cabin Data was incomplete and even

RECOMMENDATION

- Further research could investigate why male passengers had significantly lower survival rates

- Further analysis could examine why 3rd class passengers had lower survival rates, such as differences in cabin location and access to lifeboats

CONCLUSION

- Gender and pricing did factor into survival rates.
- Women were protected more resulting in women having higher rates
- When both characteristics were considered together, 1st class women had the highest survival rate while 3rd class men had one of the lowest

- Interesting how gender can play a big role in survival